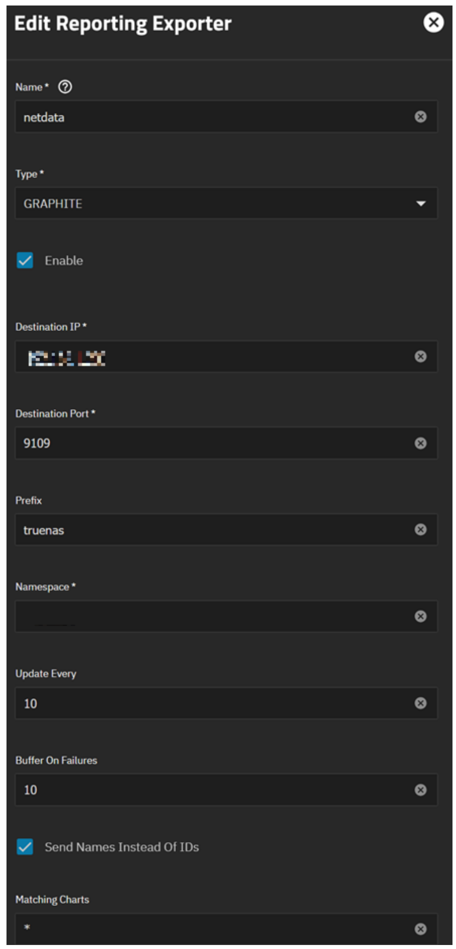
1) Create ‘Reporting Exporter’ through GRAPHITE in the Reporting tab in TrueNAS Scale:
    a. In the TrueNAS GUI, left hand navigation pane, click ‘Reporting ‘-> top right corner, click ‘Exporter’ -> click ‘Add’
    b. Fill in the field similar to this:

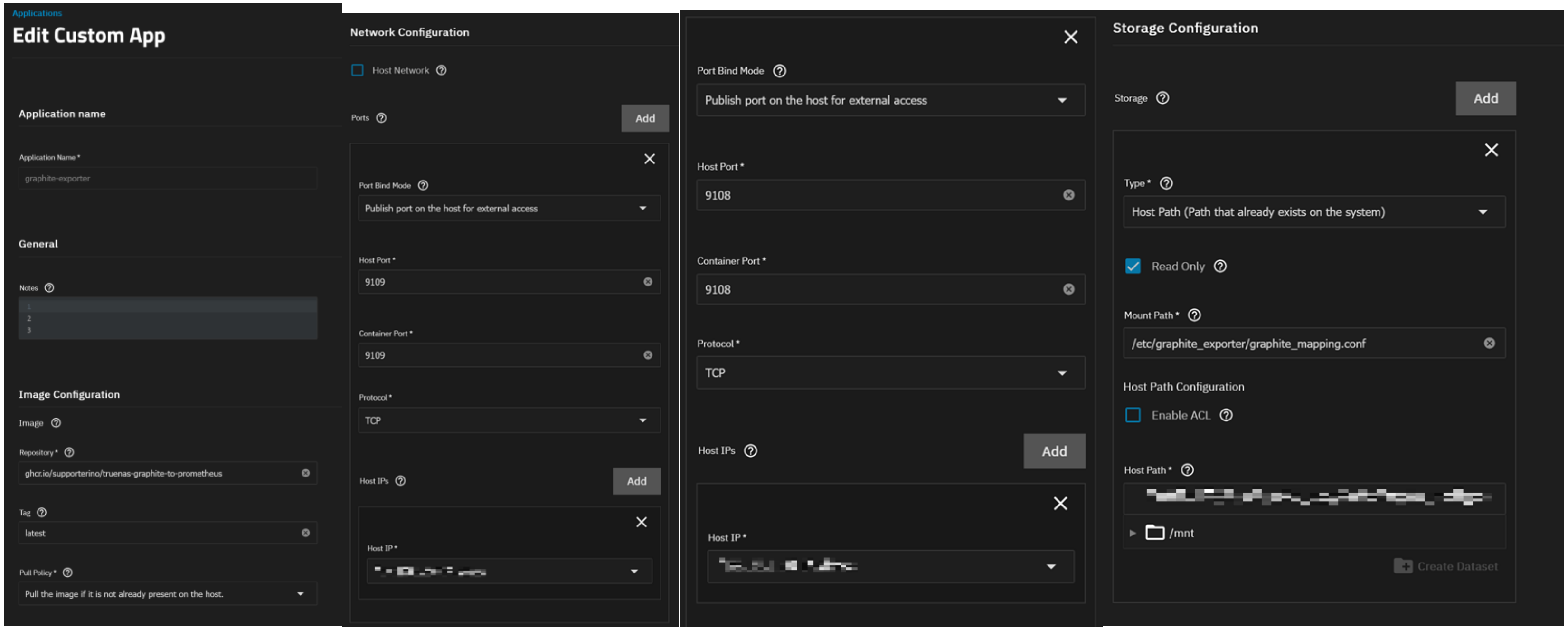


    c. Your destination IP needs to be your actual TrueNAS IP address; xxx.xxx.xxx.xxx
    d. Keep the destination port as 9109, that is how I got it to work
    e. I believe Prefix needs to be ‘truenas’ and choose another name for your Namespace, keep note of them.

2) Download ‘graphite_mapping.conf’ and ‘netdata.conf’ file from [Supporterino / truenas-graphite-to-prometheus]() (you can also follow the directions there as well).
    a. netdata.conf’ will be used in step 3) and‘graphite_mnapping.conf will be used in step 4).

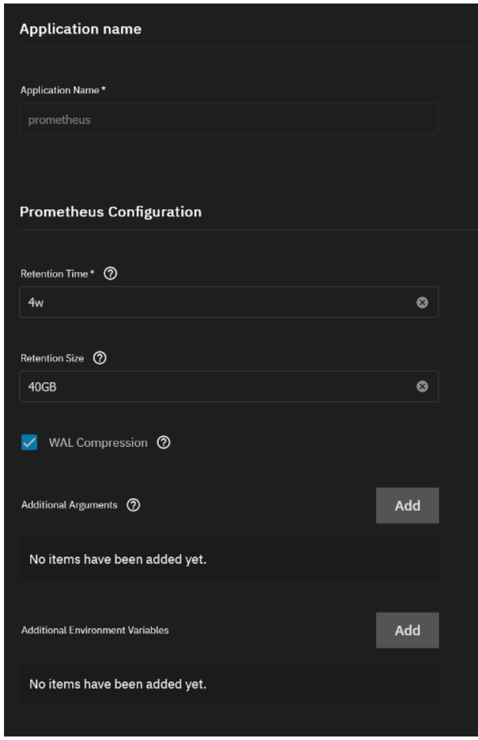
3) Copy ‘netdata/conf’ to /etc/netdata/netdata.conf
    a. Helpful commands in CLI to get it to your respective location and privilege change:
        i. `sudo cp /path/from/download/netdata.conf /etc/netdata/netdata.conf`
        ii. `sudo chown root:root /etc/netdata/netdata.conf`
        iii. `sudo chmod 644 /etc/netdata/netdata.conf`
        iv. Restart Netdata - `sudo systemctl restart netdata`
        v. Verify Netdata is running - `systemctl status netdata`

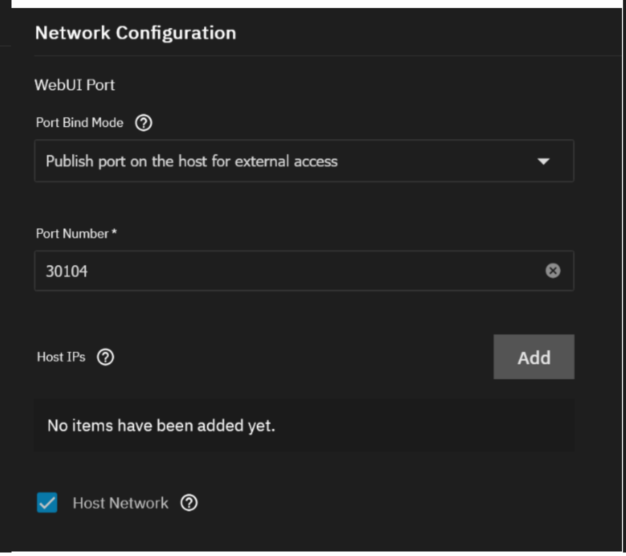
4) Install ‘graphite-exporter’ on your TrueNAS Scale as a custom app in TrueNAS GUI:
    a. Screenshots to help, in order of what you see when you are populating the information, then instructions below the images.

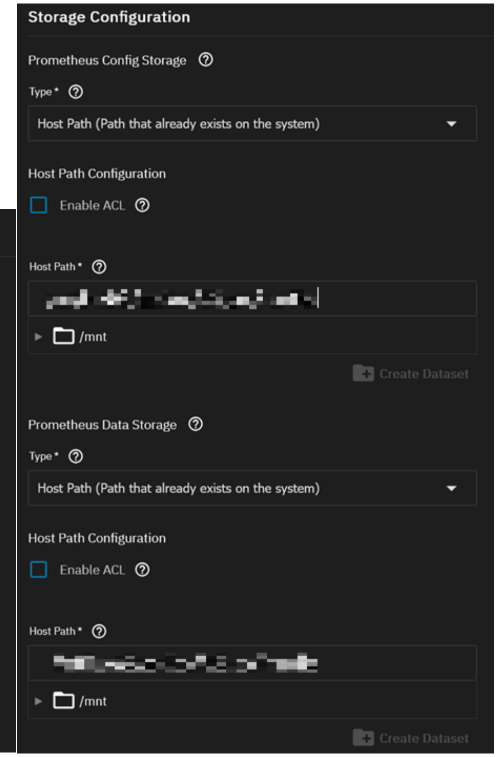

Applications
Edit Custom App
Application name
Application Name *
graphite-exporter
General
Notes
1
2
3
Image Configuration
Image
Repository *
ghcr.io/supporterino/truenas-graphite-to-prometheus
Tag
latest
Pull Policy *
Pull the image if it is not already present on the host.
Network Configuration
Host Network
Ports
Add
Port Bind Mode
Publish port on the host for external access
Host Port *
9109
Container Port *
9109
Protocol *
TCP
Host IPs
Add
Host IP *
Port Bind Mode
Publish port on the host for external access
Host Port *
9108
Container Port *
9108
Protocol *
TCP
Host IPs
Add
Host IP *
Storage Configuration
Storage
Add
Type *
Host Path (Path that already exists on the system)
Read Only
Mount Path *
/etc/graphite_exporter/graphite_mapping.conf
Host Path Configuration
Enable ACL
Host Path *
/mnt
Create Dataset

b. Left hand navigation pane, click 'Apps' -> top right corner, click 'Discover Apps' -> click 'Custom App'.
c. Fill in the following mandatory fields:
    i. Image Configuration:
        1. Image Repository: `ghcr.io/supporterino/truenas-graphite-to-prometheus`
        2. Tag: Latest
        3. Pull Policy: Pull the image if it is not already present on the host.
    ii. Network Configuration:
        1. Port Blind Mode: Publish port on the host for external access
        2. Add x1 -> Host Port: 9109; Container Port: 9109; Protocal: TCP; Host IP: Your TrueNAS IP xxx.xxx.xxx
        3. Add x2 -> Host Port: 9108; Container Port: 9108; Protocal: TCP; Host IP: Your TrueNAS IP xxx.xxx.xxx
    iii. Storage – I have a pool dedicated to Apps, so this is how I did it (your milage may vary):
        1. Type: Host Path (Path that already exists on the system)
        2. Read Only: Check
        3. Mount Path: /mnt/path/to/respective/directory/graphite_mapping.conf
            a. As noted in step 2) copy over the 'graphite_mapping.conf' file download from Supporterino / truenas-graphite-to-prometheus and place it in that repository which you will point to here.
            b. Helpful commands in CLI to get it to your respective location and privilege change:
                i. `sudo cp /path/from/download/location/graphite_mapping.conf /path/to/repo/graphite_mapping.conf`
                ii. `sudo chmod 644 /path/to/repo/graphite_mapping.conf`
    iv. Click the 'Update' button at the bottom.
d. Test in Shell -> `curl -s http://youripaddress:9108/metrics | grep truenas_cpu`
    i. You should see a list of metrics populate, your scrape is functioning

5) Install ‘prometheus’ on your TrueNAS Scale through ‘Discover Apps’:
    a. Screenshots to help, in order in what you see when you are populating the information on the next page, then instructions below the images


Application name
Application Name *
prometheus
Prometheus Configuration
Retention Time *
4w
Retention Size
40GB
WAL Compression
Additional Arguments
Add
No items have been added yet.
Additional Environment Variables
Add
No items have been added yet.


Network Configuration
WebUI Port
Port Bind Mode
Publish port on the host for external access
Port Number *
30104
Host IPs
Add
No items have been added yet.
Host Network


Storage Configuration
Prometheus Config Storage
Type *
Host Path (Path that already exists on the system)
Host Path Configuration
Enable ACL
Host Path *
/mnt
Create Dataset
Prometheus Data Storage
Type *
Host Path (Path that already exists on the system)
Host Path Configuration
Enable ACL
Host Path *
/mnt
Create Dataset

b. Left hand navigation pane, click ‘Apps’ -> top right corner, click ‘Discover Apps’
c. Search ‘Prometheus”
d. Click ‘Install’ and fill the below
   i. Application Name:
      1. Application Name: Prometheus
   ii. Prometheus Configuration:
      1. Retention Time – pick to your liking, I have 4w (w-weeks, d-days, m-months, etc.)
      2. Retention Size – pick to your liking, I have 40GB
   iii. Network Configuration:
      1. Publish port on the host for external access
      2. Port Number: 30104
   iv. Storage Configuration I have a pool dedicated to Apps, so this is how I did it (your milage may vary):
      1. Prometheus Config Storage:
         a. Type: Host Path (Path that already exists on the system)
         b. Mount Path: /mnt/path/to/respective/directory
      2. Prometheus Data Storage:
         a. Type: Host Path (Path that already exists on the system)
         b. Mount Path: /mnt/path/to/respective/directory
   v. Click the ‘Update’ button at the bottom.
   vi. I had to navigate to the promethus.yaml in Shell and update the yaml file. It did not auto-populate for some reason:
      1. Navigate to shell:
      2. Navigate to where you had your ‘Prometheus’ app installed and you should see a prometheus.yml file.
         a. Edit it by `sudo nano prometheus.yml`
         b. Leverage the below code into the file and save

```
global:
  scrape_interval: 10s
scrape_configs:
  - job_name: "truenas_graphite_exporter"
    static_configs:
      - targets: ["xxx.xxx.xxx.xxx:9108"]
        labels:
          instance: "truenas"
```

env: "xxxxx"

The "instance" and "env" should tie to step 1) where you define the "prefix" and "namespace" (red text above)

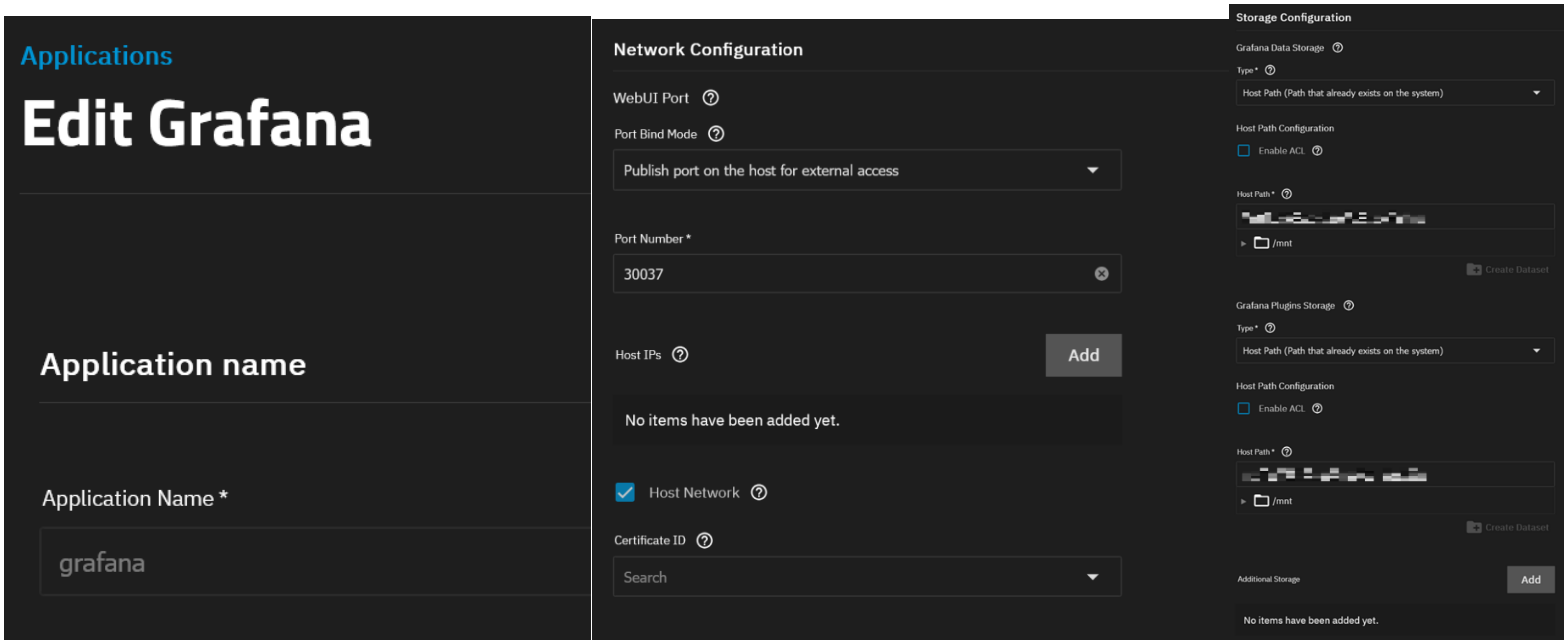
vii. Once the 'Prometheus App' status is running, click on it and Application Info side window should populate.

viii. Click on 'Web UI' -> a dialogue box should pop up, enable HTTPS Redirect, click 'Continue'

1. It should open a browser into Prometheus UI.
2. On the top banner, click Status -> then Target Health
3. In the Endpoint section, you should see your TrueNAS scale IP address
4. Click it and you can see all the data scrapes in it in text form.

6) Lastly, install 'grafana' on your TrueNAS Scale through 'Discover Apps':

a. Screenshots to help, in order in what you see when you are populating the information on the next page, then instructions below the images

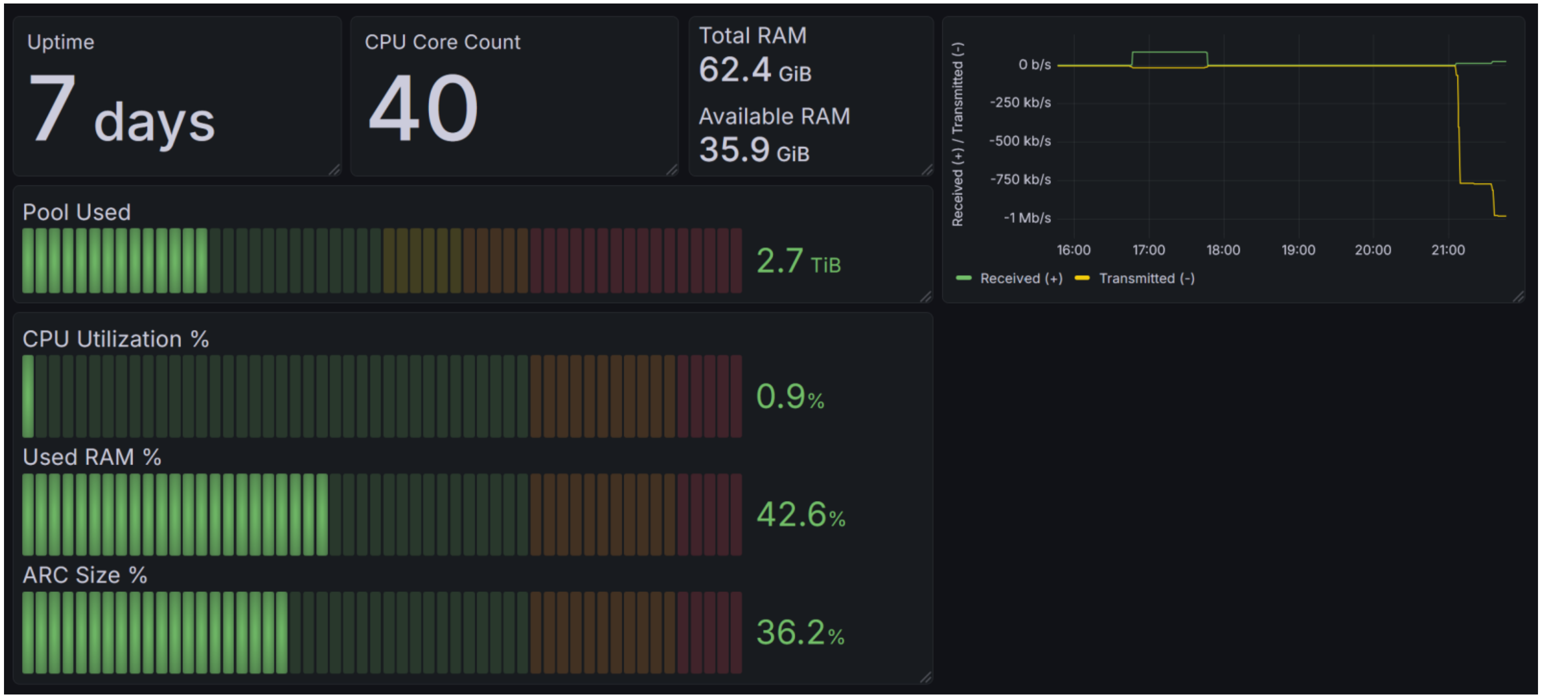


Applications
Edit Grafana
Application name
Application Name *
grafana
Network Configuration
WebUI Port
Port Bind Mode
Publish port on the host for external access
Port Number *
30037
Host IPs
Add
No items have been added yet.
Host Network
Certificate ID
Search
Storage Configuration
Grafana Data Storage
Type *
Host Path (Path that already exists on the system)
Host Path Configuration
Enable ACL
Host Path *
/mnt
Create Dataset
Grafana Plugins Storage
Type *
Host Path (Path that already exists on the system)
Host Path Configuration
Enable ACL
Host Path *
/mnt
Create Dataset
Additional Storage
Add
No items have been added yet.

b. Left hand navigation pane, click ‘Apps’ -> top right corner, click ‘Discover Apps’
c. Search ‘Grafana’
d. Click ‘Install’ and fill the below:
    i. Application Name:
        1. Application Name: Grafana
    ii. Network Configuration:
        1. Web UI
        2. Publish port on the host for external access
        3. Port Number: 30037
    iii. Storage Configuration I have a pool dedicated to Apps, so this is how I did it (your milage may vary):
        1. Grafana Data Storage
            a. Type: Host Path (Path that already exists on the system)
            b. Host Path: /mnt/path/to/respective/directory
        2. Grafana Plugins Storage:
            a. Type: Host Path (Path that already exists on the system)
            b. Host Path: /mnt/path/to/respective/directory
    iv. Click the ‘Update’ button at the bottom.
    v. Once the ‘Grafana App’ status is running, click on it and Application Info side window should populate.
    vi. Click on ‘Web UI’ -> a dialogue box should pop up, enable HTTPS Redirect, click ‘Continue’
        1. It should open a browser into Grafana UI.
        2. On the left banner, click Connections -> search Prometheus
        3. Click ‘Add new data source’ in the top right corner
        4. It should automatically find your Prometheus App
        5. You can then navigate to ‘Dashboards’ in the left banner and click ‘New’ in the top right corner -> ‘New Dashboard’.
        6. It should then automatically suggest the Prometheus data source for you to select.

You should now be set to create dashboards and begin monitoring your TrueNAS SCALE system.

Here is a sample of what I have created, the below dashboard image is still work in progress:


Uptime
7 days
CPU Core Count
40
Total RAM
62.4 GiB
Available RAM
35.9 GiB
Received (+) / Transmitted (-)
0 b/s
-250 kb/s
-500 kb/s
-750 kb/s
-1 Mb/s
16:00
17:00
18:00
19:00
20:00
21:00
Received (+)
Transmitted (-)
Pool Used
2.7 TiB
CPU Utilization %
0.9%
Used RAM %
42.6%
ARC Size %
36.2%